
author: The Tuan Trinh -- date: 2026-05-11

```
In [1]: import os
import platform

print("=" * 100)
print("OS INFORMATION")
print("=" * 100)

print("OS Name:", os.name)
print("System:", platform.system())
print("Release:", platform.release())
print("Version:", platform.version())
print("Machine:", platform.machine())
print("Processor:", platform.processor())
print("Platform:", platform.platform())

print("\nEnvironment Variables (sample):")
for key, value in list(os.environ.items())[:10]:
    print(f"{key} = {value}")
```

```
=====
OS INFORMATION
=====
```

```
OS Name: posix
System: Linux
Release: 4.18.0-348.12.2.el8_5.x86_64
Version: #1 SMP Wed Jan 19 14:35:04 EST 2022
Machine: x86_64
Processor: x86_64
Platform: Linux-4.18.0-348.12.2.el8_5.x86_64-x86_64-with-glibc2.28
```

```
Environment Variables (sample):
```

```
BASH_FUNC__module_raw%% = () { eval `usr/bin/tclsh /usr/share/Modules/libexec/modulecmd.tcl bash "$@"`;
    _mlstatus=$?;
    return $_mlstatus
}
BASH_FUNC_ml%% = () { module ml "$@"
}
BASH_FUNC_module%% = () { _module_raw "$@" 2>&1
}
BASH_FUNC_scl%% = () { if [ "$1" = "load" -o "$1" = "unload" ]; then
    eval "module $@";
else
    /usr/bin/scl "$@";
fi
}
BASH_FUNC_which%% = () { ( alias;
    eval ${which_declare} ) | /usr/bin/which --tty-only --read-alias --read-functions --show-tilde --show-dot "$@"
}
CHROME_DESKTOP = code.desktop
COLORTERM = truecolor
CONDA_DEFAULT_ENV = RadarLM
CONDA_EXE = /work/tuan.tt19010226/miniconda/bin/conda
CONDA_PREFIX = /work/tuan.tt19010226/conda/envs/RadarLM
```

```
In [2]: platform.python_version()
```

```
Out[2]: '3.10.19'
```

Lab1 - Basic Operations on Images

Basic Operations on Images

Goal Learn to:

Access pixel values and modify them Access image properties Set a Region of Interest (ROI) Split and merge images
Almost all the operations in this section are mainly related to Numpy rather than OpenCV. A good knowledge of Numpy is required to write better optimized code with OpenCV.

Accessing and Modifying pixel values. Let's load a color image first:

```
In [3]: import cv2
import urllib.request
import numpy as np
import matplotlib.pyplot as plt
```

```
url = "https://lh3.googleusercontent.com/sitesv/AA5AbUBqcFI1QBKtKz0ggbZc1K1wDc5YvsfzoaTS4tbDqFk5k8qKh8xhRgtzG70:
resp = urllib.request.urlopen(url)
img = np.asarray(bytearray(resp.read()), dtype=np.uint8)
img = cv2.imdecode(img, cv2.IMREAD_COLOR)
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
plt.imshow(img_rgb)
```

Out[3]: <matplotlib.image.AxesImage at 0x14e85f98c7c0>



In [4]: !pip install opencv-python

Requirement already satisfied: opencv-python in ./conda/envs/RadarLM/lib/python3.10/site-packages (4.13.0.92)
Requirement already satisfied: numpy>=2 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from opencv-python) (2.2.6)

In [5]: px = img[100,100]
print(px)
blue = img[100,100,0]
print(blue)

[189 207 217]
189

In [6]: img[100,100] = [255,255,255]
print(img[100,100])

[255 255 255]

Accessing Image Properties

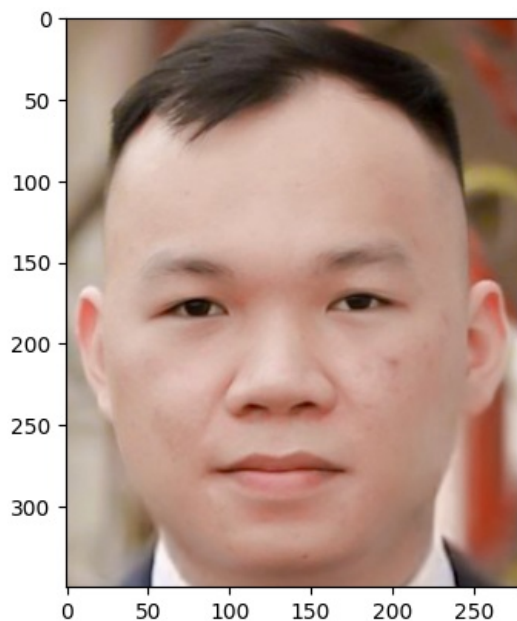
In [7]: print("Shape of Image: ", img.shape)
print("Size of Image: ", img.size)
print("Image datatype: ", img.dtype)

Shape of Image: (1723, 1280, 3)
Size of Image: 6616320
Image datatype: uint8

Image ROI

In [8]: face = img[300:650, 466:745]
face_rgb = cv2.cvtColor(face, cv2.COLOR_BGR2RGB)
plt.imshow(face_rgb)

Out[8]: <matplotlib.image.AxesImage at 0x14e854cbfb80>



Splitting and Merging Image Channels

```
In [9]: b,g,r = cv2.split(img)
```

Making Borders for Images (Padding)

```
In [10]: # url_ = "https://lh3.googleusercontent.com/sitesv/AA5AbUAR7zNH3a_XV05nKR7Xwj3WZAqyg5NCtAp80gdiy01APjmo58tdqqsBi
url_ = "https://th.bing.com/th/id/ODF.yf34kah7rom5d3lzVLg1Cw?w=32&h=32&qlt=90&pcl=ffffffa&o=6&pid=1.2"

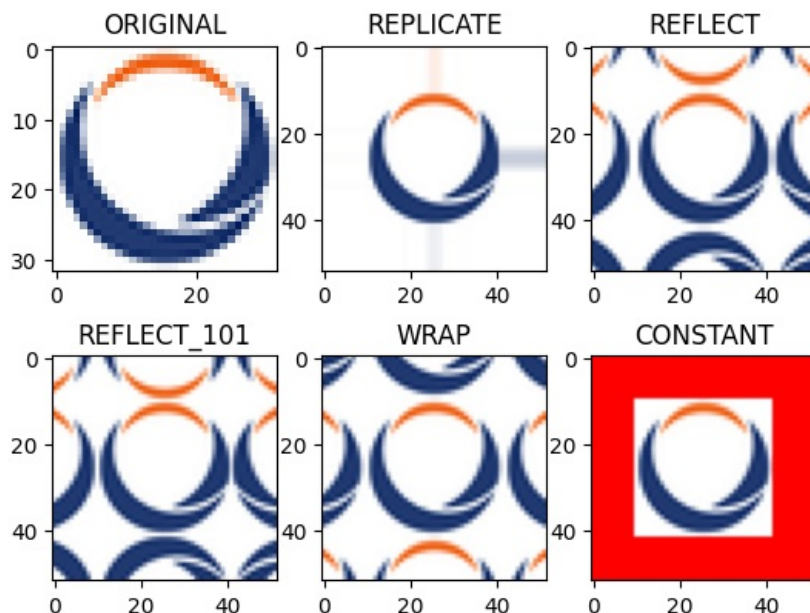
# Download image
resp = urllib.request.urlopen(url_)
image_ = np.asarray(bytearray(resp.read()), dtype=np.uint8)
img_1 = cv2.imdecode(image_, cv2.IMREAD_COLOR)
img_1 = cv2.cvtColor(img_1, cv2.COLOR_BGR2RGB)
```

```
In [11]: BLUE = [255,0,0]

replicate = cv2.copyMakeBorder(img_1,10,10,10,10,cv2.BORDER_REPLICATE)
reflect = cv2.copyMakeBorder(img_1,10,10,10,10,cv2.BORDER_REFLECT)
reflect101 = cv2.copyMakeBorder(img_1,10,10,10,10,cv2.BORDER_REFLECT_101)
wrap = cv2.copyMakeBorder(img_1,10,10,10,10,cv2.BORDER_WRAP)
constant= cv2.copyMakeBorder(img_1,10,10,10,10,cv2.BORDER_CONSTANT,value=BLUE)

plt.subplot(231),plt.imshow(img_1,'gray'),plt.title('ORIGINAL')
plt.subplot(232),plt.imshow(replicate,'gray'),plt.title('REPLICATE')
plt.subplot(233),plt.imshow(reflect,'gray'),plt.title('REFLECT')
plt.subplot(234),plt.imshow(reflect101,'gray'),plt.title('REFLECT_101')
plt.subplot(235),plt.imshow(wrap,'gray'),plt.title('WRAP')
plt.subplot(236),plt.imshow(constant,'gray'),plt.title('CONSTANT')

plt.show()
```



```
In [12]: !pip install notebook
```

```
Requirement already satisfied: notebook in ./conda/envs/RadarLM/lib/python3.10/site-packages (7.5.6)
Requirement already satisfied: jupyter-server<3,>=2.4.0 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from
notebook) (2.18.2)
Requirement already satisfied: jupyterlab-server<3,>=2.28.0 in ./conda/envs/RadarLM/lib/python3.10/site-packages
(from notebook) (2.28.0)
Requirement already satisfied: jupyterlab<4.6,>=4.5.7 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from
notebook) (4.5.7)
Requirement already satisfied: notebook-shim<0.3,>=0.2 in ./conda/envs/RadarLM/lib/python3.10/site-packages (fro
m notebook) (0.2.4)
Requirement already satisfied: tornado>=6.2.0 in /home/tuan.tt19010226/.local/lib/python3.10/site-packages (from
notebook) (6.5.1)
Requirement already satisfied: anyio>=3.1.0 in /home/tuan.tt19010226/.local/lib/python3.10/site-packages (from j
upyter-server<3,>=2.4.0->notebook) (4.11.0)
Requirement already satisfied: argon2-cffi>=21.1 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jupy
ter-server<3,>=2.4.0->notebook) (25.1.0)
Requirement already satisfied: jinja2>=3.0.3 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jupyter-
server<3,>=2.4.0->notebook) (3.1.6)
Requirement already satisfied: jupyter-client>=7.4.4 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from
jupyter-server<3,>=2.4.0->notebook) (8.8.0)
Requirement already satisfied: jupyter-core!=5.0.*,>=4.12 in ./conda/envs/RadarLM/lib/python3.10/site-packages (
from jupyter-server<3,>=2.4.0->notebook) (5.9.1)
Requirement already satisfied: jupyter-events>=0.11.0 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from
jupyter-server<3,>=2.4.0->notebook) (0.12.1)
Requirement already satisfied: jupyter-server-terminals>=0.4.4 in ./conda/envs/RadarLM/lib/python3.10/site-packa
ges (from jupyter-server<3,>=2.4.0->notebook) (0.5.4)
Requirement already satisfied: nbconvert>=6.4.4 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jupyt
er-server<3,>=2.4.0->notebook) (7.17.1)
Requirement already satisfied: nbformat>=5.3.0 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jupyte
r-server<3,>=2.4.0->notebook) (5.10.4)
Requirement already satisfied: overrides>=5.0 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jupyter
-server<3,>=2.4.0->notebook) (7.7.0)
Requirement already satisfied: packaging>=22.0 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jupyt
er-server<3,>=2.4.0->notebook) (26.2)
Requirement already satisfied: prometheus-client>=0.9 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from
jupyter-server<3,>=2.4.0->notebook) (0.25.0)
Requirement already satisfied: pyzmq>=24 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jupyter-serv
er<3,>=2.4.0->notebook) (27.1.0)
Requirement already satisfied: send2trash>=1.8.2 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jupy
ter-server<3,>=2.4.0->notebook) (2.1.0)
Requirement already satisfied: terminado>=0.8.3 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jupyt
er-server<3,>=2.4.0->notebook) (0.18.1)
Requirement already satisfied: traitlets>=5.6.0 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jupyt
er-server<3,>=2.4.0->notebook) (5.15.0)
Requirement already satisfied: websocket-client>=1.7 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from
jupyter-server<3,>=2.4.0->notebook) (1.9.0)
Requirement already satisfied: async-lru>=1.0.0 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jupyt
erlab<4.6,>=4.5.7->notebook) (2.3.0)
Requirement already satisfied: httpx<1,>=0.25.0 in /home/tuan.tt19010226/.local/lib/python3.10/site-packages (fr
om jupyterlab<4.6,>=4.5.7->notebook) (0.28.1)
Requirement already satisfied: ipykernel!=6.30.0,>=6.5.0 in ./conda/envs/RadarLM/lib/python3.10/site-packages (f
rom jupyterlab<4.6,>=4.5.7->notebook) (7.2.0)
Requirement already satisfied: jupyter-lsp>=2.0.0 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jup
yterlab<4.6,>=4.5.7->notebook) (2.3.1)
Requirement already satisfied: setuptools>=41.1.0 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jup
yterlab<4.6,>=4.5.7->notebook) (80.9.0)
Requirement already satisfied: tomli>=1.2.2 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jupyterla
```

b<4.6,>=4.5.7->notebook) (2.4.1)
Requirement already satisfied: certifi in ./conda/envs/RadarLM/lib/python3.10/site-packages (from httpx<1,>=0.25.0->jupyterlab<4.6,>=4.5.7->notebook) (2026.4.22)
Requirement already satisfied: httpcore==1.* in /home/tuan.tt19010226/.local/lib/python3.10/site-packages (from httpx<1,>=0.25.0->jupyterlab<4.6,>=4.5.7->notebook) (1.0.9)
Requirement already satisfied: idna in ./conda/envs/RadarLM/lib/python3.10/site-packages (from httpx<1,>=0.25.0->jupyterlab<4.6,>=4.5.7->notebook) (3.14)
Requirement already satisfied: h11>=0.16 in /home/tuan.tt19010226/.local/lib/python3.10/site-packages (from httpcore==1.*->httpx<1,>=0.25.0->jupyterlab<4.6,>=4.5.7->notebook) (0.16.0)
Requirement already satisfied: babel>=2.10 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jupyterlab-server<3,>=2.28.0->notebook) (2.18.0)
Requirement already satisfied: json5>=0.9.0 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jupyterlab-server<3,>=2.28.0->notebook) (0.14.0)
Requirement already satisfied: jsonschema>=4.18.0 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jupyterlab-server<3,>=2.28.0->notebook) (4.26.0)
Requirement already satisfied: requests>=2.31 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jupyterlab-server<3,>=2.28.0->notebook) (2.33.1)
Requirement already satisfied: exceptiongroup>=1.0.2 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from anyio>=3.1.0->jupyter-server<3,>=2.4.0->notebook) (1.3.1)
Requirement already satisfied: sniffio>=1.1 in /home/tuan.tt19010226/.local/lib/python3.10/site-packages (from anyio>=3.1.0->jupyter-server<3,>=2.4.0->notebook) (1.3.1)
Requirement already satisfied: typing_extensions>=4.5 in /home/tuan.tt19010226/.local/lib/python3.10/site-packages (from anyio>=3.1.0->jupyter-server<3,>=2.4.0->notebook) (4.15.0)
Requirement already satisfied: argon2-cffi-bindings in ./conda/envs/RadarLM/lib/python3.10/site-packages (from argon2-cffi>=21.1->jupyter-server<3,>=2.4.0->notebook) (25.1.0)
Requirement already satisfied: comm>=0.1.1 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.7->notebook) (0.2.3)
Requirement already satisfied: debugpy>=1.6.5 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.7->notebook) (1.8.20)
Requirement already satisfied: ipython>=7.23.1 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.7->notebook) (8.37.0)
Requirement already satisfied: matplotlib-inline>=0.1 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.7->notebook) (0.2.2)
Requirement already satisfied: nest-asyncio>=1.4 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.7->notebook) (1.6.0)
Requirement already satisfied: psutil>=5.7 in /home/tuan.tt19010226/.local/lib/python3.10/site-packages (from ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.7->notebook) (7.0.0)
Requirement already satisfied: decorator in /home/tuan.tt19010226/.local/lib/python3.10/site-packages (from ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.7->notebook) (5.2.1)
Requirement already satisfied: jedi>=0.16 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.7->notebook) (0.19.2)
Requirement already satisfied: pexpect>4.3 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.7->notebook) (4.9.0)
Requirement already satisfied: prompt_toolkit<3.1.0,>=3.0.41 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.7->notebook) (3.0.52)
Requirement already satisfied: pygments>=2.4.0 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.7->notebook) (2.20.0)
Requirement already satisfied: stack_data in ./conda/envs/RadarLM/lib/python3.10/site-packages (from ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.7->notebook) (0.6.3)
Requirement already satisfied: wcwidth in ./conda/envs/RadarLM/lib/python3.10/site-packages (from prompt_toolkit<3.1.0,>=3.0.41->ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.7->notebook) (0.7.0)
Requirement already satisfied: parso<0.9.0,>=0.8.4 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jedi>=0.16->ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.7->notebook) (0.8.7)
Requirement already satisfied: MarkupSafe>=2.0 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jinja2>=3.0.3->jupyter-server<3,>=2.4.0->notebook) (2.1.5)
Requirement already satisfied: attrs>=22.2.0 in /home/tuan.tt19010226/.local/lib/python3.10/site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.28.0->notebook) (25.3.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.28.0->notebook) (2025.9.1)
Requirement already satisfied: referencing>=0.28.4 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.28.0->notebook) (0.37.0)
Requirement already satisfied: rpds-py>=0.25.0 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.28.0->notebook) (0.30.0)
Requirement already satisfied: python-dateutil>=2.8.2 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jupyter-client>=7.4.4->jupyter-server<3,>=2.4.0->notebook) (2.9.0.post0)
Requirement already satisfied: platformdirs>=2.5 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jupyter-core!=5.0.*,>=4.12->jupyter-server<3,>=2.4.0->notebook) (4.9.6)
Requirement already satisfied: python-json-logger>=2.0.4 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook) (4.1.0)
Requirement already satisfied: pyyaml>=5.3 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook) (6.0.3)
Requirement already satisfied: rfc3339-validator in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook) (0.1.4)
Requirement already satisfied: rfc3986-validator>=0.1.1 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook) (0.1.1)
Requirement already satisfied: fqdn in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook) (1.5.1)
Requirement already satisfied: isoduration in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook) (20.11.0)
Requirement already satisfied: jsonpointer>1.13 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook) (3.1.1)
Requirement already satisfied: rfc3987-syntax>=1.1.0 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook) (1.1.0)

Requirement already satisfied: uri-template in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook) (1.3.0)

Requirement already satisfied: webcolors>=24.6.0 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook) (25.10.0)

Requirement already satisfied: beautifulsoup4 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook) (4.14.3)

Requirement already satisfied: bleach!=5.0.0 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from bleach[css]!=5.0.0->nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook) (6.3.0)

Requirement already satisfied: defusedxml in ./conda/envs/RadarLM/lib/python3.10/site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook) (0.7.1)

Requirement already satisfied: jupyterlab-pygments in ./conda/envs/RadarLM/lib/python3.10/site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook) (0.3.0)

Requirement already satisfied: mistune<4,>=2.0.3 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook) (3.2.1)

Requirement already satisfied: nbclient>=0.5.0 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook) (0.10.4)

Requirement already satisfied: pandocfilters>=1.4.1 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook) (1.5.1)

Requirement already satisfied: webencodings in ./conda/envs/RadarLM/lib/python3.10/site-packages (from bleach!=5.0.0->bleach[css]!=5.0.0->nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook) (0.5.1)

Requirement already satisfied: tinycss2<1.5,>=1.1.0 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from bleach[css]!=5.0.0->nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook) (1.4.0)

Requirement already satisfied: fastjsonschema>=2.15 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from nbformat>=5.3.0->jupyter-server<3,>=2.4.0->notebook) (2.21.2)

Requirement already satisfied: ptyprocess>=0.5 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from pexpect>4.3->ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.7->notebook) (0.7.0)

Requirement already satisfied: six>=1.5 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from python-dateutil>=2.8.2->jupyter-client>=7.4.4->jupyter-server<3,>=2.4.0->notebook) (1.17.0)

Requirement already satisfied: charset-normalizer<4,>=2 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from requests>=2.31->jupyterlab-server<3,>=2.28.0->notebook) (3.4.7)

Requirement already satisfied: urllib3<3,>=1.26 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from requests>=2.31->jupyterlab-server<3,>=2.28.0->notebook) (2.7.0)

Requirement already satisfied: lark>=1.2.2 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from rfc3987-syntax>=1.1.0->jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook) (1.3.1)

Requirement already satisfied: cffi>=1.0.1 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from argon2-cffi-bindings->argon2-cffi>=21.1->jupyter-server<3,>=2.4.0->notebook) (2.0.0)

Requirement already satisfied: pycparser in ./conda/envs/RadarLM/lib/python3.10/site-packages (from cffi>=1.0.1->argon2-cffi-bindings->argon2-cffi>=21.1->jupyter-server<3,>=2.4.0->notebook) (3.0)

Requirement already satisfied: soupsieve>=1.6.1 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from beautifulsoup4->nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook) (2.8.3)

Requirement already satisfied: arrow>=0.15.0 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from isoduration->jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook) (1.4.0)

Requirement already satisfied: tzdata in ./conda/envs/RadarLM/lib/python3.10/site-packages (from arrow>=0.15.0->isoduration->jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook) (2026.2)

Requirement already satisfied: executing>=1.2.0 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from stack_data->ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.7->notebook) (2.2.1)

Requirement already satisfied: asttokens>=2.1.0 in ./conda/envs/RadarLM/lib/python3.10/site-packages (from stack_data->ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.7->notebook) (3.0.1)

Requirement already satisfied: pure_eval in ./conda/envs/RadarLM/lib/python3.10/site-packages (from stack_data->ipython>=7.23.1->ipykernel!=6.30.0,>=6.5.0->jupyterlab<4.6,>=4.5.7->notebook) (0.2.3)